AIGC 5500 | MIDTERM PROJECT | 25% OF FINAL GRADE

Midterm Project: Comparing Convolutional, Recurrent and Attention-Based Models

Due date posted on Blackboard

Over the first five modules you met three architecture families: convolutional networks, recurrent networks, and attention-based models. In this project your group implements one model from each family, trains all three on the same task under identical conditions, and compares them against a simple baseline. You then remove or change one component of each model to see what it contributes, check whether your findings hold on a second dataset, and explain what you found in a short report and a recorded video presentation. What is being assessed is the quality of your reasoning and evidence, not how high your accuracy is. A modest result that is tested honestly beats an impressive result that cannot be trusted.

What you will do

1. **Choose a task that all three families can handle.** Any public dataset that a free Colab or Kaggle GPU session can train on several times within your time budget, and that can be presented to a convolutional, a recurrent and an attention-based model. Choose one where a simple baseline does not already reach 99%, because then there is nothing left to compare.
2. **Build a defensible baseline.** A simpler model that your three models should beat, such as logistic regression or a small multilayer perceptron on the same inputs, trained on the same split, with the same tuning effort and the same training budget. A baseline you did not tune is not defensible.
3. **Implement one model from each family.** A convolutional model, a recurrent model (a plain RNN, an LSTM or a GRU) and an attention-based model (a Transformer encoder, or a vision-style model over patches). Write the model code yourselves in PyTorch and train from scratch. You may use standard layers such as nn.Linear, nn.Conv2d or nn.LSTM. Pretrained weights are not allowed, because they would make the comparison depend on the pretraining, not on the architecture.
4. **Compare them under identical conditions.** The same split, the same optimizer family, the same training budget and the same tuning effort for all models, with parameter counts of a similar size (within about a factor of two of each other, or the difference reported and explained).
5. **Run one ablation per family.** For each of your three models, remove or change exactly one component, hold everything else fixed, and measure the difference.
6. **Check a second dataset.** Train the baseline and your three models on a different dataset and ask whether the ranking and your main findings still hold.
7. **Measure what each model costs.** Report the parameter count, the inference time per sample on stated hardware, and peak memory during inference, for every model.
8. **Report and present.** A written report and a recorded video presentation, described below.

Choosing a task

The table below shows how one task can be given to each family. These are examples to show the shape, not a menu. You are encouraged to choose your own task, as long as all three families can be applied to it in a fair way.

Task	Convolutional	Recurrent	Attention-based
Small image classification	A CNN over the pixels.	An LSTM or GRU that reads the image one row at a time.	A Transformer encoder over image patches.
Time series classification, such as activity recognition from sensor windows	A 1D CNN along the time axis.	An LSTM or GRU over the time steps.	A Transformer encoder over the time steps, with positions.
Short text classification on a small vocabulary	A 1D CNN over word embeddings.	An LSTM or GRU over the words.	A Transformer encoder over the words, with positions.

Keep the scope realistic. Every variant must be trained several times, so a single run should take minutes, not hours. Two other courses in your program, AIGC 5501 and AIGC 5502, cover natural language processing and image processing in depth. Your project should not be a pipeline-building exercise for either of those. Its subject is the comparison between architectures.

Scope

Every group delivers the same scope. The group divides the work into five pieces: the baseline and the shared evaluation code, the convolutional model with its ablation, the recurrent model with its ablation, the attention-based model with its ablation, and the second dataset check. Every member owns at least one of these pieces. The owner designs it, runs it, explains it in the report, and can explain and defend it if asked, although every member should still understand the whole project.

Piece	What it involves
Core comparison	The baseline and the three models on one dataset, three seeds each, under identical conditions, with the cost of each model measured.
Three ablations	One ablation per family. Each changes exactly one component of that family’s model.
Second dataset check	The baseline and the three models, three seeds each, on a different dataset.

The table below gives examples of ablations. You may choose others.

Family	Example ablations
Convolutional	Remove the skip connections, remove batch normalization, or change the kernel size.
Recurrent	Plain RNN against LSTM, gate bias initialization, or unidirectional against bidirectional.
Attention-based	Remove positions, change the number of heads, or compare pre-norm with post-norm.

The second dataset check asks whether your findings hold somewhere else. A ranking that appears on one dataset and reverses on another is a result worth reporting.

Requirements

Your submission must do all of the following. Each is checked directly during marking.

- Split your data into training, validation and test sets before doing anything else. Use the validation set to make every decision. Evaluate on the test set once, at the end, for the final models only.
- Train the baseline, all three models and every ablation variant under the same conditions: the same split, the same optimizer family, the same training budget, and the same tuning effort. State what you tuned and how much for each.

- Report every seed, not only the mean, with at least three seeds for the baseline, each model and each ablation variant, and on the second dataset. Report the mean and the spread.
- Make your experiments reproducible: set and report seeds, pin versions in requirements.txt, and keep the code that produced each number in the repository. An experiment described only in prose, without the code and output that produced it, receives a significantly reduced experimentation mark.
- Keep the numbers consistent. Figures, tables and prose must agree. An internal contradiction is marked down in every criterion it touches.
- Report the inference cost of every model: parameter count, time per sample, and peak memory, with the hardware named.
- Explain the differences between families using what the course taught about how each one behaves, for example local patterns, a sequential state, or a direct lookup. A table of scores without a mechanism is not an explanation.
- Report negative and surprising results honestly. An ablation that shows a component does not help, or a family that loses to the baseline, is a valid and valuable result if it is explained.
- Cite anything you did not write, including code from tutorials, documentation, forums and AI assistants.

Watch out

Using the test set to choose between models, even once, makes the reported test result an optimistic estimate and is treated as a serious methodological error. Decide everything on the validation set. If you look at test performance and then change your model, you have used the test set for selection. The test split exists to be touched once.

Why a baseline, three families and ablations

They answer different questions. A baseline answers whether a deep model is worth using at all. Comparing the three families answers which way of building in structure suits your task, and why. An ablation answers which part of a model is responsible for its result. A project with strong scores but no baseline cannot say whether depth mattered, and a project with a comparison but no ablation cannot say why one family did better.

Deliverables

You submit one thing: the URL of your GitHub repository. Everything else is delivered inside the repository, and nothing is uploaded to Blackboard.

Item in the repository	Requirement
GitHub repository	The URL is the submission. The repository is shared with the instructor account below on the day you start.
Runnable notebook	A .ipynb that runs from a clean environment and reproduces every table and figure in the report. A PDF report without the notebook does not earn full marks on this deliverable.
Python modules	The code the notebook imports, such as the model definitions and the training and evaluation code, as .py files in the repository.
Report	A PDF following the outline below, at most eight pages.
Slides	The slides used in the video, saved as a PDF.
Video presentation	A recording of 15 to 20 minutes by one member, uploaded to Humber OneDrive and shared as described below. The README contains the link.

Report outline

Use these headings. The report should be written so that someone who has not seen your notebook can judge your claims.

1. **Problem and data.** Your task, the dataset, the split sizes and the class balance.
2. **The three models and why.** How each model is built, how the data is presented to it, and the reasons for each design choice, tied to what the course taught about that family.
3. **Baseline.** What it is, why it is a fair comparison, and what you tuned.
4. **Results.** A table with every seed for the baseline and the three models, the mean and spread, and the single final test result for each, followed by an explanation of the differences between families.
5. **Ablations.** For each of the three ablations, what you changed, a table with every seed, and what the difference shows.
6. **Second dataset.** The dataset, why it was chosen, a table with every seed, and whether your findings still hold.
7. **Inference cost.** Parameter count, time per sample, peak memory, and the hardware, for every model.
8. **Limitations.** What your evidence does not show: the datasets, the budget, the number of seeds, and anything that could change the conclusion.
9. **Contributions and references.** Who owned which piece, and a list of everything you did not write.

Do not leave template placeholders, such as bracketed prompts or sample text, in the submitted PDF. Each one is flagged.

Video presentation

Your group records one video of the project, uploads it to Humber OneDrive, and links it from the README. One member presents the work, and that member is named in the README. The other members do not need to appear in the video, but every member should be able to explain any part of the project, including the code, if asked.

The video must meet all of the following. Each is checked directly during marking.

- **Length.** Between 15 and 20 minutes. A video shorter than 15 minutes or longer than 20 minutes is penalized, and the instructor is not required to watch beyond 20 minutes. Rehearse with a timer before you record.
- **One presenter.** One member speaks throughout, in their own voice, over the slides. Do not use a synthesized voice.
- **Structure.** The task and the question, the three models, the baseline and the results, the ablations, the second dataset, the cost, and what you would not claim from your evidence.
- **Evidence on screen.** Show the results table with every seed, and spend about thirty seconds on the notebook output that produced it, so the instructor can see that the numbers come from your code.
- **Anticipated questions.** Finish by raising two or three questions that a sceptical reader would ask, for example whether the baseline was tuned fairly or whether an ablation changed only one thing, and answer them from your evidence.
- **Slides.** Show the slides full screen. They must be readable at 720p and carry no more text than you speak about. The same slides are committed to the repository as a PDF.
- **Audio and format.** Clear audio, no background music, recorded at 720p or higher and saved as MP4. Play the file from start to finish before you upload it.
- **Upload and sharing.** Upload the video to the OneDrive of your Humber account. Select Share, choose the option for people in Humber with the link, and set the permission to view only. Also share it directly with hossein.pourmodheji@humber.ca. Test the link in a private browser window, signing in with a different Humber account if you can, to confirm that it plays without a request for access.
- **Link in the README.** Put the link under a heading called Video presentation, with the presenter's name and the length of the video. Upload the video before the due date posted on Blackboard, and do not move, rename, replace or delete it after you submit.

A video that the instructor cannot open is treated as not submitted.

Commit history

Share the repository with the instructor account on the day you start. Your commit history is part of your mark. Work in commits, with messages that describe what changed, throughout the project. A commit history concentrated near the deadline, or a single commit made the night before, is a hard deduction. Every member must have commits under their own account, and the README contribution table names the owner of each ablation.

Repository requirements

Item	Requirement
.gitignore	Present, committed before your first code commit, and excluding .venv, __pycache__, .env, and any large data or checkpoint files
README.md	The question, the dataset and how to obtain it, how to run the notebook, a results summary, a contribution table, and the link to the video
requirements.txt	Generated with pip freeze, including torch and matplotlib
Notebook and Python modules	The runnable .ipynb and the .py modules it imports, with seeds set and reported
Commit history	At least twelve commits spread over at least five different days, each a coherent change with a descriptive message, with at least two from every member

How marks are awarded

The project is worth 25% of your final grade. A group that delivers less than the scope above loses marks under the experiments criterion.

Component	Weight
Implementation and experiments: the three models, the baseline, the ablations and the repository	15%
Report: results, consistency, cost, limitations and justification	5%
Video presentation: timing, clarity and the evidence behind your claims	5%

How to submit

Post the URL of your GitHub repository to the Midterm Project submission on Blackboard. Do not upload files, and do not submit a link to a single file within the repository. One submission per group. The link to your video goes in the README, not in a separate submission.

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https://github.com/your-username/your-repository
```

Make sure the repository is public. If you prefer to keep it private, you must share it with the instructor account below. Sharing with any other account does not count as a submission and will be rejected.

Instructor GitHub account

`hossein.pourmodheji@humber.ca`

Add this account as a collaborator under Settings, Collaborators in your repository. Any other account will be rejected.

A repository that the instructor cannot open is treated as not submitted, and the late policy applies from the point at which access is granted.

Before you submit, run this check

Clone your own repository into a different folder, create a fresh virtual environment, install from requirements.txt, and run the notebook from the top. Check that every table and figure in the report matches the notebook output, that no placeholder text remains in the PDF, and that your video runs between 15 and 20 minutes and its link opens in a private browser window. If everything agrees, your submission is complete. If it does not, you have found the problem before the instructor did.